SAYAN CHAKRABARTY

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Department of Statistics
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Personal Website
Google Scholar

RESEARCH INTERESTS

Statistical methods for complex datasets, interpretable machine learning for heterogeneous relational data, scalable inference for large networks, and biomedical applications of statistics and machine learning.

EDUCATION

University of Illinois Urbana-Champaign — Champaign, IL
PhD in Statistics — 2024
Advisors: Yuguo Chen and Srijan Sengupta
Collaborators: Fatima Husain and Raksha Mudar
Thesis: Subsampling Based Inference for Network Data

Indian Statistical Institute — Kolkata, India
MSTAT with Specialization in Computational Statistics Track — 2018

St. Xavier's College — Kolkata, India
BSc (Honors) in Statistics with Minors in Mathematics and Computer Science — 2016

POSITIONS

Department of Statistics, University of Michigan — Ann Arbor, MI
Postdoctoral Research Fellow in Statistics — 2024 – Present
Advisors: Elizaveta Levina and Ji Zhu
Collaborators: Snigdha Panigrahi, Shelly Flagel and Akbar Waljee

Department of Statistics, University of Illinois Urbana-Champaign — Champaign, IL
Graduate Teaching / Research Assistant — 2018 – 2024

Illinois Statistics Office, University of Illinois Urbana-Champaign — Champaign, IL
Statistical Consultant — 2020 – 2024
Provided statistical consultation on different stages of scientific experiments to clients from both academia and industry. Collaborated on research projects in medicine, biology, psychology, geology, finance, law, linguistics, engineering, and political science.

Information Security and Data Protection Unit, Synchrony Financial — Champaign, IL (Remote)
Summer Intern — Summer 2021
Analyzed and predicted the number of data breaches and compared the efficacies of multiple data protection measures.

CHEGG INC. — Kolkata, India
Tutor of Statistics and Probability (Part time) — 2014 – 2016

PUBLICATIONS

Published / Accepted

Chakrabarty, S., Sengupta, S., Chen, Y. (2026). Network Cross-Validation and Model Selection via Subsampling. *In Press at Journal of the Royal Statistical Society Series B, arXiv preprint:2504.06903.* Preprint.

Chakrabarty, S., Sengupta, S., Chen, Y. (2025). Subsampling Based Community Detection for Large Networks. *Statistica Sinica* (35) 1627 – 1648. Article.

Moran, C., Park, T., Buffenstein, R., **Chakrabarty, S.**, Lindeblad, M., Fortman, J., Adams, C. (2024) Pharmacokinetics of Injectable Meloxicam and Buprenorphine in the Naked Mole-Rat (Heterocephalus glaber). *Journal of the American Association for Laboratory Animal Science* (63) 565 – 571. Article

Chakrabarty, S., Mudar, R., Chen, Y., Husain, F. (2024). Contribution of Tinnitus and Hearing Loss to Depression: NHANES Population Study. *Ear and Hearing* (45) 775 – 786. Article

Han, M. D., Kwon, T. G., Miloro, M., **Chakrabarty, S.** (2023). What Is the Linear Accuracy of Regional Voxel-Based Registration for Orthognathic Surgery Landmarks? *Journal of Oral and Maxillofacial Surgery* (81) 546 – 556. Article

Ghosh, A., SahaRay, R., **Chakrabarty, S.**, Bhadra, S. (2021). Robust Generalised Quadratic Discriminant Analysis. *Pattern Recognition* (117) 107981. Article

In Preparation

Chakrabarty, S., Levina, E. Resampling Networks and Network Statistics.

Chakrabarty, S., Levina, E. Fast Network Subsampling and Aggregation Using Overlapping Partitions.

Chakrabarty, S., Panigrahi, S., Levina, E. Randomized Network Cross-Validation.

Chakrabarty, S., Levina, E., Zhu, J. Local Feature Importance in Network Linked Machine Learning Prediction Models.

Ramos, B., **Chakrabarty, S.**, Cain, C., Levina, E., Kennedy, R., Flagel, S. Identifying a Neurochemical Signature of the Paraventricular Nucleus of the Thalamus That Predicts Individual Variability in Cue-Reward Learning.

Hou, J., **Chakrabarty, S.**, Huang, J., Van, T., Baker, J., Zhu, J., Waljee, A. Machine Learning Models to Predict Objective Remission in Veterans with Ulcerative Colitis Treated with Thiopurines.

Teaching Experience

Department of Statistics, University of Illinois Urbana-Champaign — Champaign, IL

Instructor

STAT 200: Introduction to Statistical Analysis — Fall 2019

Delivered classroom instruction and held office hours for a section of approximately 40 freshmen and sophomores from multiple disciplines. The course included fundamental pre-calculus statistical concepts and methods.

Discussion Leader

STAT 400: Statistics and Probability I — Fall 2018, Spring and Summer 2019

Led classroom discussion sessions, held office hours and graded for 4 sections of 25 junior and senior statistics major students per term. The course included topics on probability, random variables, basic inference and sampling distributions.

Teaching Assistant

Duties were mentoring class projects, holding office hours and grading assignments.

STAT427: Statistical Consulting — Springs 2021, 2022, 2023

STAT578: Doctoral Topic Course - Quantile Regression — Fall 2022

STAT511: Advanced Mathematical Statistics — Fall 2021

STAT430: Graduate Topic Course - Deep Learning — Fall 2020

STAT578: Doctoral Topic Course - Advanced Bayesian Analysis — Spring 2020

Department of Mathematics, University of Illinois Urbana-Champaign — Champaign, IL

Workshop Co-Host and Student Project Mentor

Computation and Data Science Summer Boot Camp — Summer 2019

Prepared workshop materials, hosted sessions on R and Python for Data Science and mentored student projects for a 2 week summer camp training approximately 30 mathematics doctoral students in data analysis and programming.

Awards and Achievements

Travel Award, *Recent Advances in Random Networks, Institute of Mathematical and Statistical Innovation* — 2026

Travel Award, *International New Researchers' Conference, Institute of Mathematical Statistics* — 2025

Travel Award, *International Indian Statistical Association Conference* — 2025

Travel Award, *Symposium on New Frontiers in Reliability and Risk Analysis: A Tribute to Nozer D. Singpurwalla, George Washington University* — 2023

Awarded 'Excellent in Teaching', *University of Illinois Urbana-Champaign* — 2019

All India Rank 8 in Mathematical Statistics at *Indian Institute of Technology Joint Admission Test for Masters* — 2016

INSPIRE SHE Fellowship, *Department of Science & Technology, Govt. of India* — 2013 – 2018

PRESENTATIONS

The 9th International Conference on Econometrics and Statistics (EcoSta) — Tokyo, Japan
Invited Talk: Bootstrapping Network Statistics Using Overlapping Partitions — 2026

International Symposium on Nonparametric Statistics — Thessaloniki, Greece
Invited Talk: Network Bootstrap Using Overlapping Partitions — 2026

Recent Advances in Random Networks — Chicago, IL
Lightning Talk: Network Bootstrap Using Overlapping Partitions — 2026
Poster: Network Cross-Validation and Model Selection via Subsampling

International Conference on Statistics and Data Science — Seville, Spain
Contributed Talk: Network Bootstrap Using Overlapping Partitions — 2025

International Conference on Computational and Mathematical Statistics — London, UK
Invited Talk: Network Bootstrap Using Overlapping Partitions — 2025

International New Researchers' Conference — Seville, Spain
Poster: Network Cross-Validation and Model Selection via Subsampling — 2025

Joint Statistical Meetings — Nashville, TN
Contributed Talk: Network Bootstrap Using Overlapping Partitions — 2025

International Indian Statistical Association Conference — Lincoln, NE
Invited Talk: Network Bootstrap Using Overlapping Partitions — 2025

Department of Statistics, North Carolina State University — Raleigh, NC
Course Presentation: Network Corss-Validation and Model Selection via Subsampling — 2025

The Michigan Student Symposium for Interdisciplinary Statistical Sciences — Ann Arbor, MI
Contributed Talk: Network Bootstrap Using Overlapping Partitions — 2025

Symposium on New Frontiers in Reliability and Risk Analysis — Washington, DC
Invited Poster: Subsampling Based Model Selection for Network Data — 2023

Joint Statistical Meetings — Toronto, Canada
Contributed Talk: Subsampling Based Model Selection for Network Data — 2023

International Indian Statistical Association Conference — Golden, CO
Student Poster: Subsampling Based Community Detection for Large Networks — 2023

Joint Statistical Meetings — Washington, DC
Invited Talk and Poster: Subsampling Based Community Detection for Large Networks — 2022

Institute of Mathematical Statistics Annual Meeting — London, UK
Contributed Talk: Subsampling Based Community Detection for Large Networks — 2022

New England Statistics Symposium — Storrs, CO
Invited Student Talk: Subsampling Based Community Detection for Large Networks — 2022

Bohrer Workshop — Champaign, IL
Student Talk and Poster: Subsampling Based Community Detection for Large Networks — 2022

PROFESSIONAL SERVICE

Journal Reviewer — 2024 – Present
Journal of American Statistical Association
Computational Statistics and Data Analysis
Journal of Computational and Graphical Statistics
Journal of the Royal Statistical Society: Series B

University Service
Postdoctoral Representative to the Department of Statistics Cultural Committee, University of Michigan — 2025 – 2026
Founding Chair of the Doctoral Student Seminar Series at the Department of Statistics, University of Illinois Urbana-Champaign — 2022
Member of the Statistics Doctoral Student Association at the Department of Statistics, University of Illinois Urbana-Champaign — 2021 – 2023
Founding Member of the PhD Student Seminar Series at the Department of Statistics, University of Illinois Urbana-Champaign — 2019 – 2020

Volunteer Work
Membership Committee, International Indian Statistical Association — 2025 – Present

SOFTWARE

netOP: R package for network operations using overlapping partitions. GitHub

COMPUTING SKILLS

Programming Languages: R, Rcpp, Python, SPSS, SAS, C/C++

Miscellaneous: Git/GitHub, Linux, HPC, Docker/Singularity